Assignment 1 - Project Plan Document (BRD)

Business Requirements Document (BRD)

Project Name: Agroledger

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1. Introduction

1.1 Purpose:

The purpose of this Business Requirements Document (BRD) is to outline the business requirements for developing a blockchain-based supply chain traceability solution for the Canadian Sheep Federation (CSF). The chosen technology platform for this project is Hyperledger Fabric, which provides a secure and permissioned blockchain environment.

1.2 Background:

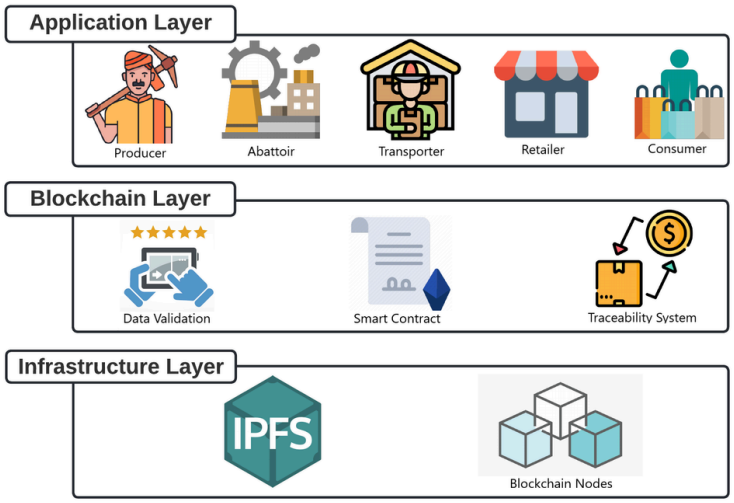
The current supply chain management system used by the CSF relies on traditional methods that are susceptible to inefficiencies and errors. This project aims to introduce a blockchain-based solution using Hyperledger Fabric to ensure secure, transparent, and efficient data management across the entire sheep supply chain.

1.3 Scope:

The project will involve developing a blockchain-based system that tracks the lifecycle of sheep from birth to slaughter, ensuring transparency, security, and efficiency. This includes the implementation of smart contracts, data recording, access control, and stakeholder-specific permissions.

Diagram 1 - High-Level Architecture Diagram

Purpose: This diagram will provide a visual overview of the entire system architecture, showing how the blockchain network, data storage layers, integration layer, user interface, and API gateway interact within the solution.



2. Project Objectives

2.1 Primary Objectives:

- To achieve complete traceability of sheep from birth to carcass disposal using blockchain technology.
- To ensure data integrity and prevent unauthorized data modifications across the supply chain.

2.2 Secondary Objectives:

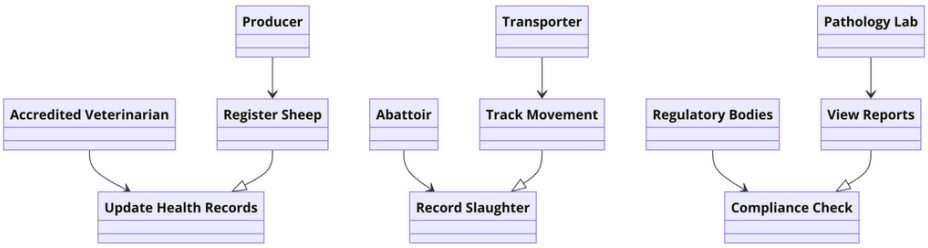
- To enhance efficiency in data entry and retrieval for all stakeholders.
- To implement secure, role-based access to data for different stakeholders.

2.3 Long-Term Vision:

To establish a trusted platform that provides end consumers, regulators, and international partners with the ability to verify the origin and authenticity of sheep products.

Diagram 2 - Use Case Diagram

Purpose: To provide an overview of all functions or "use cases" the system provides and the stakeholders involved. This helps to identify the necessary system capabilities and their users.



3. Stakeholders

3.1 List of Stakeholders:

- Abattoir
- Accredited Veterinarian
- Assembly Yard
- Auction Livestock
- Industry Representatives
- Pathology Lab
- Producer
- Program Administrator
- Research Facility
- Responsible Administrator
- Feedlot
- Tag Dealer

- Tag Distribution
- Tag Manufacturer
- Test Station
- Transporter
- Distributor
- Retailer
- Restaurateur
- Consumer

3.2 Roles and Responsibilities:

- **Abattoir:** Records data related to slaughter and carcass evaluation.
- **Accredited Veterinarian:** Updates health records and certification statuses.
- **Assembly Yard:** Manages sheep collection and distribution data.
- **Auction Livestock:** Manages data related to sheep auctions and sales.
- **Industry Representatives:** Ensures compliance with industry standards.
- **Pathology Lab:** Provides health and disease data.
- **Producer:** Manages data on sheep breeding, health, and movement.
- **Program Administrator:** Oversees system compliance and data integrity.
- **Research Facility:** Conducts research on sheep genetics and manages related data.
- **Responsible Administrator:** Handles administrative data and system oversight.
- **Feedlot:** Manages feeding and health data.
- **Tag Dealer, Tag Distribution, Tag Manufacturer:** Manages tag data from manufacturing to distribution.
- **Test Station:** Conducts genetic and health tests on sheep.
- **Transporter:** Manages transportation data for sheep between facilities.
- **Distributor:** Manages the distribution and logistics data for further stages in the supply chain.
- **Retailer:** Manages data related to sheep product retailing.
- **Restaurateur:** Manages data related to sheep product use in the food service industry.
- **Consumer:** Views product origin and traceability information to verify authenticity.

4. Project Scope

4.1 In-Scope Activities:

- Development and deployment of a blockchain-based system using Hyperledger Fabric.
- Implementation of smart contracts for automated processes.
- Integration of IoT devices for real-time monitoring of sheep during transportation.

4.2 Out-of-Scope Activities:

- Development of new IoT hardware devices.
- Interoperability integration with international supply chain systems beyond the initial scope of Canadian operations.

5. Functional Requirements

5.1 Data Recording and Tracking:

The system must record all critical events such as birth, health checks, feeding, movement, and slaughter of sheep.

Diagram 3 - Sequence Diagrams for Stakeholder Interactions

Purpose: Include sequence diagrams for each key event (Sheep Registration, Sheep Movement Tracking, Slaughter and Carcass Disposal). These diagrams illustrate the interactions between stakeholders (e.g., Producer, Transporter, Abattoir) and the blockchain system during critical events in the sheep supply chain.

Diagram 3.1 - Sheep Registration

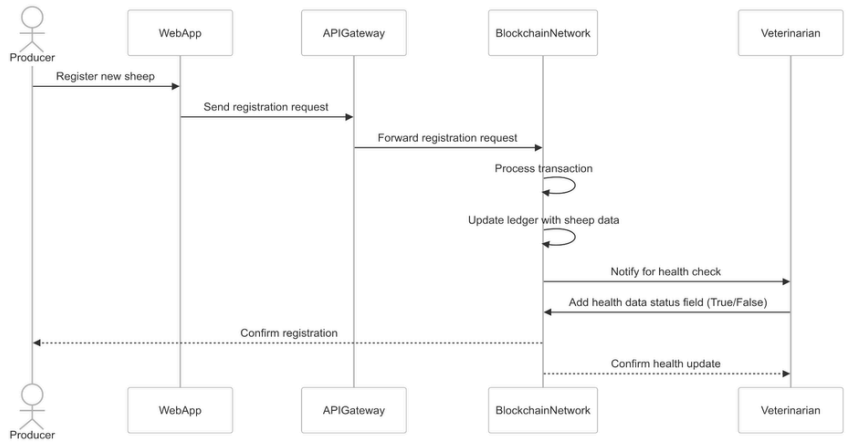


Diagram 3.2 - Sheep Movement Tracking

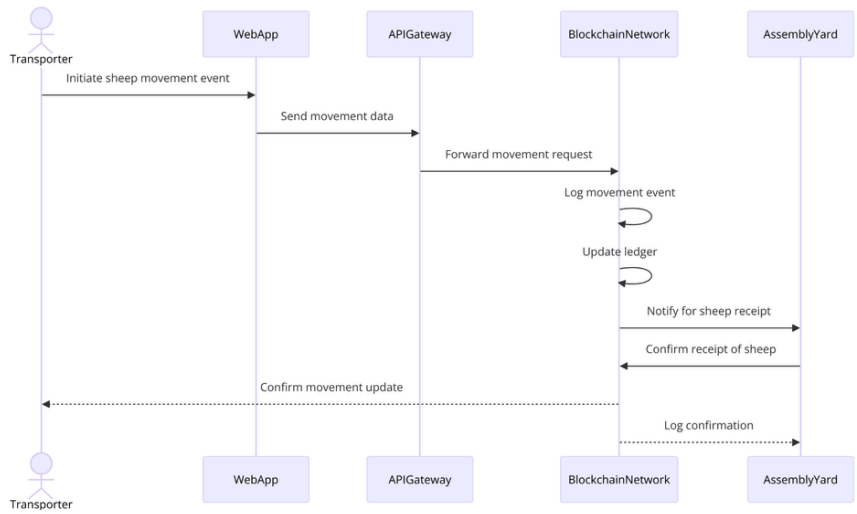
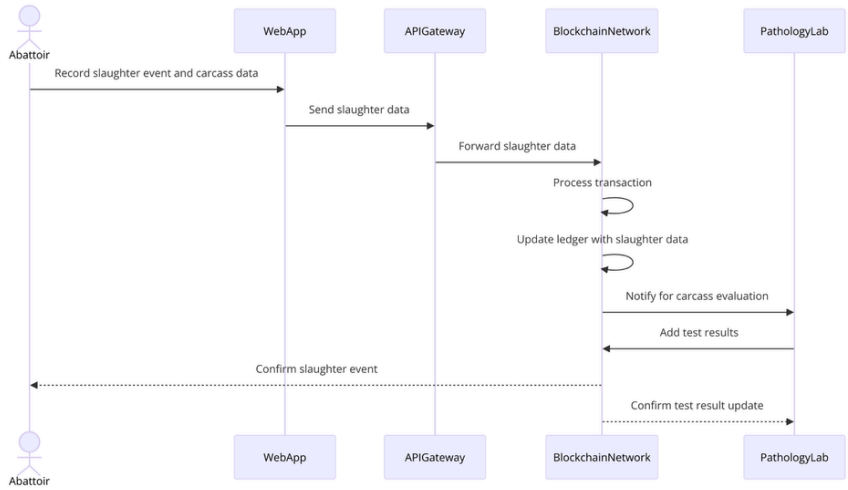


Diagram 3.3 - Slaughter and Carcass Disposal

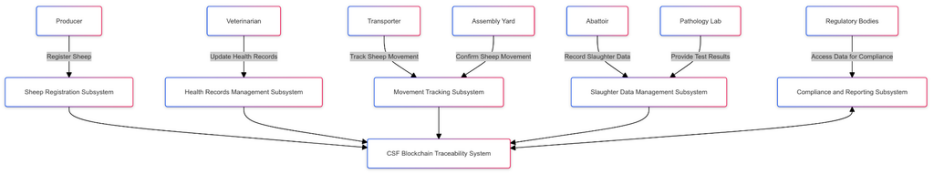


5.2 Data Access Control:

Role-based access control (RBAC) will be implemented to manage data access based on stakeholder roles.

Diagram 4 - Data Flow Diagram (DFD)

Purpose: This diagram shows the flow of data within the system, illustrating how data moves between different components and stakeholders.



5.3 Event Handling and Tracking:

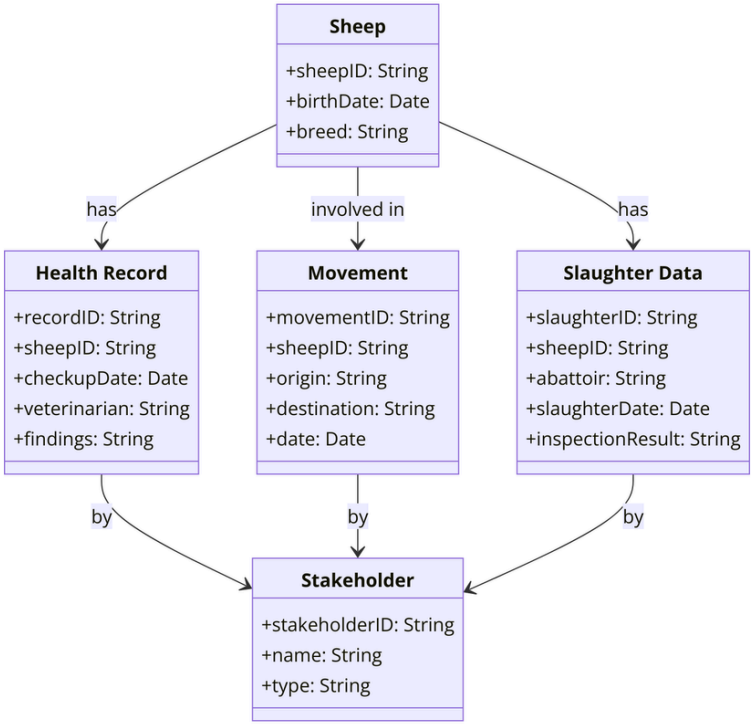
Automation of event recordings to maintain accuracy and timeliness in the supply chain.

5.4 Smart Contract Development:

Smart contracts will be developed to handle data validation, access control, and automation of critical supply chain processes.

Diagram 5 - Entity-Relationship Diagram (ERD)

Purpose: To define the database schema and ensure all data points are captured accurately. It illustrates the relationships between different entities such as Sheep, Health Record, Movement, Slaughter Data, and Stakeholders.



6. Non-Functional Requirements

6.1 Security:

Data must be encrypted both at rest and in transit to prevent unauthorized access.

6.2 Performance:

The system should support high volumes of transactions without degradation in performance.

6.3 Scalability:

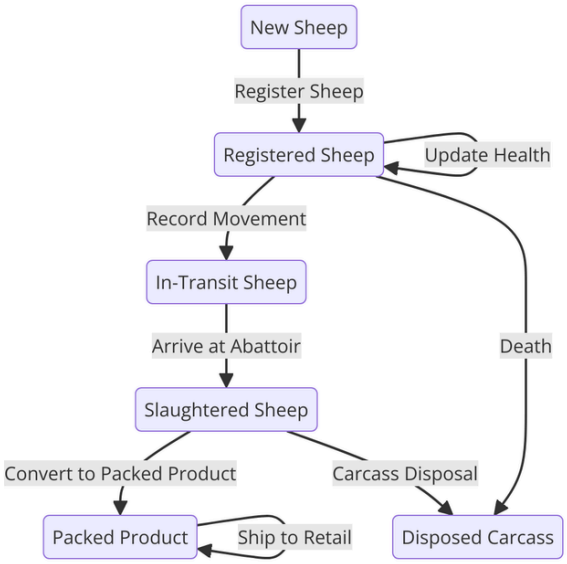
The solution must be scalable to accommodate a growing number of stakeholders and data entries.

6.4 Usability:

The user interface should be intuitive and easy to use for all stakeholders.

Diagram 6 - State Transition Diagram

Purpose: Illustrates the states of the system or components and the transitions between those states based on events, such as "Register Sheep," "Update Health," "Record Movement," "Perform Slaughter."



7. Technology Platform

7.1 Blockchain Platform: Hyperledger Fabric

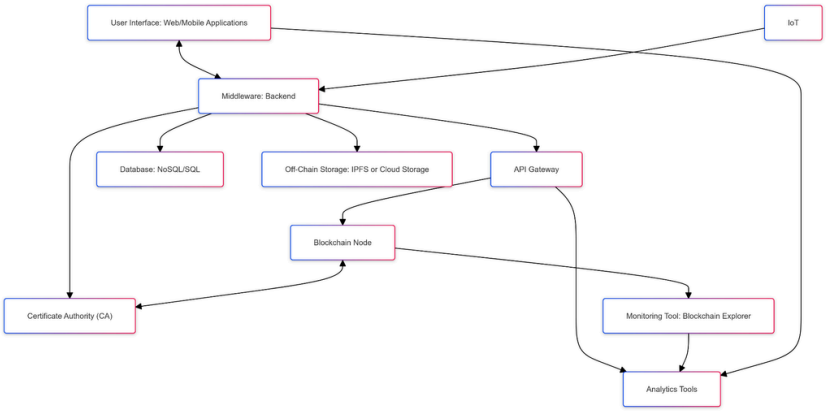
Hyperledger Fabric is chosen for its permissioned capabilities, allowing controlled access to data and ensuring data integrity and immutability.

7.2 Technical Specifications:

- Network Configuration: Multiple peers and orderers will manage transactions and consensus.
- Certificate Authority (CA) for identity management and Membership Service Providers (MSPs) for access control.
- **Technology Stack**
  - Blockchain: Hyperledger Fabric
  - Backend: Node.js with Express.js
  - Frontend: React
  - Database: MySQL
  - Version Control: GitHub

Diagram 7 - Component Diagram

Purpose: Provides a detailed view of the software and hardware components, focusing on their interactions and interfaces, helping to visualize the architecture in terms of component



8. Permissions Matrix

8.1 Stakeholder Access Levels:

Stakeholder	Birth Record	Health Check	Movement Data	Slaughter Data	Carcass Evaluation	Genetic Testing	Tagging Data	Feedlot Data	Transport Data	Distribution Data	Retail Data	Consumption Data

Abattoir	Read	None	None	Write	Write	None	None	None	None	None	None	None
Accredited Veterinarian	Read	Write	None	None	None	Write	None	None	None	None	None	None
Assembly Yard	None	None	Write	None	None	None	None	None	Write	None	None	None
Auction Livestock	None	None	None	None	None	None	None	None	None	None	None	None
Industry Representatives	Read	Read	Read	Read	Read	Read	Read	Read	Read	Read	Read	Read
Pathology Lab	None	Write	None	Read	Write	None	None	None	None	None	None	None
Producer	Write	Read	Write	Read	None	Read	Read	Read	Read	None	None	None
Program Administrator	Read	Read	Read	Read	Read	Read	Read	Read	Read	Read	Read	Read
Research Facility	Read	Read	Read	None	None	Write	Read	Read	None	None	None	None
Responsible Administrator	Read	Read	Read	Read	Read	Read	Read	Read	Read	Read	Read	Read
Feedlot	None	None	None	None	None	None	None	Write	None	None	None	None
Tag Dealer	None	None	None	None	None	None	Write	None	None	None	None	None
Tag Distribution	None	None	None	None	None	None	Write	None	None	None	None	None
Tag Manufacturer	None	None	None	None	None	None	Write	None	None	None	None	None
Test Station	Read	Write	None	None	None	Write	None	None	None	None	None	None
Transporter	None	None	Write	None	None	None	None	None	Write	None	None	None

Distributor	None	None	None	None	None	None	None	None	Write	Write	None	None
Retailer	None	None	None	None	None	None	None	None	None	Read	Write	None
Restaurateur	None	None	None	None	None	None	None	None	None	None	Read	Write
Consumer	None	None	None	None	None	None	None	None	None	None	None	Read

8.2 Event Access and Permissions:

- **Write:** Stakeholder can enter or modify data.
- **Read:** Stakeholder can view data but cannot modify it.
- **None:** Stakeholder has no access to this data.

9. Regulatory Compliance

9.1 CFIA Standards:

The solution must comply with Canadian Food Inspection Agency (CFIA) standards to ensure data integrity and traceability.

9.2 Data Privacy Regulations:

All data handling must adhere to data privacy laws, including secure storage, access, and sharing of stakeholder information.

10. Assumptions and Constraints

10.1 Assumptions:

- All stakeholders will have access to the necessary technology to interact with the blockchain system.
- Sufficient funding and resources are available for the project's full implementation and ongoing support.

10.2 Constraints:

- The solution must comply with existing regulatory requirements and standards set by CFIA and other relevant authorities.
- The project is limited by the current IT infrastructure capabilities of the CSF and the stakeholders involved.

11. Risks and Mitigations

11.1 Risk Analysis:

- **Data Security Breach:** Potential unauthorized access or tampering with sensitive data.
- **Stakeholder Resistance:** Possible resistance from stakeholders unfamiliar with blockchain technology.

11.2 Mitigation Strategies:

- **Data Security Breach:** Implement robust encryption methods.
- **Stakeholder Resistance:** Provide comprehensive training and support to ensure stakeholders are comfortable using the new system.

12. Preferred Communication Channels

- **Primary Communication:** Jira, Slack, Github, Email, Confluence for documentation, code and updates.
  - **Jira**: For task management and tracking progress.
  - **Slack**: For quick communication and team collaboration.
  - **GitHub**: For code version control and repository management.
  - **Email**: For formal communication and sharing documents.
  - **Confluence**: For documentation and knowledge sharing.
  - **Zoom**: For virtual meetings, discussions, and presentations.
- **Secondary Communication:** Scheduled Zoom meetings for discussions and clarifications.

13. Conclusion

13.1 Summary:

This BRD outlines the comprehensive requirements for developing a blockchain-based supply chain traceability solution using Hyperledger Fabric for CSF in this POC (Proof of Concept). The project aims to enhance data integrity, transparency, and efficiency across the sheep supply chain in Canada.

14. Appendices

A. Glossary of Terms

- **Hyperledger Fabric:** A blockchain framework implementation and one of the Hyperledger projects hosted by The Linux Foundation.
- **Smart Contract:** A self-executing contract with the terms directly written into code.

B. References

- Canadian Food Inspection Agency (CFIA) standards documentation.
- Hyperledger Fabric official documentation.
- AgroLedger User Events Permissions Excel Sheet provided by CSF

Roadmap overview

AgroLedger

Client must be authenticated to access this resource.